



Customer Data Transformation & Feature Engineering Project

Python

Pandas

Scikit-Learn

Status



Project Overview

This project demonstrates a complete **Data Transformation, Data Cleaning, Feature Engineering, Outlier Detection, Feature Scaling, and Dataset Preparation Pipeline** using a customer registration dataset containing 200 users.

The objective is to transform raw customer data into a machine-learning-ready dataset while maintaining data quality and generating useful business insights.



Dataset Information

Original Dataset

Feature	Description
user_id	Unique User ID
name	Customer Name
age	Customer Age
gender	Gender Category
city	Customer City
registration_date	Registration Date

Dataset Size

- Records: **200**

- Features: 6



Technologies Used

- Python
- Pandas
- NumPy
- Scikit-Learn
- SciPy
- SQLite3
- JSON
- YData Profiling

◆ Task 1: Load Dataset

```
csv_df = pd.read_csv("users.csv")  
  
print(csv_df.head())
```

Output

user_id	name	age	gender	city	registration_date
U0001	Vihaan Sharma	35	Other	Jaipur	2022-09-08
U0002	Sai Reddy	30	Other	Hyderabad	2023-11-24
U0003	Aarohi Gupta	37	Other	Indore	2022-02-02
U0004	Aarav Gupta	44	Male	Kolkata	2023-06-02
U0005	Sara Sharma	30	Other	Chennai	2024-01-04

Screenshot

◆ Task 2: Dataset Information

```
csv_df.info()
```

Output Summary

- Total Records = 200
- Total Features = 6
- Missing Values = 0
- Duplicate Records = 0

Screenshot

◆ Task 3: Missing Value Analysis

```
csv_df.isnull().sum()
```

Result

Column	Missing Values
user_id	0
name	0
age	0
gender	0
city	0
registration_date	0

Screenshot

◆ Task 4: JSON Conversion

```
csv_df.to_json(  
    "user_dataset.json",  
    orient="records",  
    indent=4  
)
```

Output

Successfully converted CSV dataset into JSON format.

Screenshot

◆ Task 5: SQLite Database Storage

```
csv_df.to_sql(  
    "users",  
    conn,  
    if_exists="replace",  
    index=False  
)
```

Output

Dataset stored successfully in SQLite Database.

Screenshot

◆ Task 6: Missing Value Imputation

Mean Imputation

```
SimpleImputer(strategy='mean')
```

Result:

```
Missing Values After Imputation = 0
```

Most Frequent Imputation

```
SimpleImputer(strategy='most_frequent')
```

Result:

```
gender = 0  
city = 0
```

KNN Imputation

```
KNNImputer(n_neighbors=3)
```

Screenshot

◆ Task 7: Duplicate Handling

```
csv_df.drop_duplicates()
```

Result

```
Before : (200,6)  
After  : (200,6)
```

Screenshot

◆ Task 8: Outlier Detection

Z-Score Method

```
zscore()
```

Result:

```
Outliers Found = 0
```

IQR Method

Detected Outliers:

- Sai Singh (Age 51)
- Reyansh Verma (Age 53)

Screenshot

◆ Task 9: Winsorization

```
winsorize()
```

Output

age	age_winsor
35	35
30	30
37	37

Screenshot

◆ Task 10: Date Transformation

```
csv_df['Day']  
csv_df['Month']  
csv_df['Year']
```

New Features Created

- Day
- Month
- Year

Screenshot

◆ Task 11: Encoding Techniques

Label Encoding

```
Yes → 1  
No → 0
```

One Hot Encoding

Applied on:

- city
- gender

Ordinal Encoding

```
Silver → 0  
Gold → 1  
Platinum → 2
```

Screenshot

◆ Task 12: Binning

Age Segmentation

Age	Group
0-25	Low
26-40	Medium
41+	High

Spending Segmentation

Spending	Group
Low	
Medium	
High	

Screenshot

◆ Task 13: Data Transformation

Log Transformation

```
np.log1p(age)
```

Square Root Transformation

```
np.sqrt(age)
```

Screenshot

◆ Task 14: Feature Scaling

StandardScaler

```
mean = 0  
std = 1
```

MinMaxScaler

```
range = 0 → 1
```

Comparison

Method	Range
StandardScaler	(-3 to +3)
MinMaxScaler	0 to 1

Screenshot

◆ Task 15: Feature Engineering

Average Monthly Spend

```
groupby(['user_id', 'month'])
```

Purchase Frequency

```
groupby('user_id').size()
```

Days Since Last Purchase

```
today - purchase_date
```

Category-wise Expenditure

```
groupby('category')
```

Screenshot

◆ Task 16: Final Dataset Preparation

Final Dataset Shape

```
Rows      : 200  
Columns   : 20
```

Features Created

14 New Features

Screenshot



Final Report

Metric	Value
Records Before Cleaning	200
Records After Cleaning	200
Original Features	6
Features Created	14
Total Features	20
Missing Values Before	0
Missing Values After	0
Outliers Before	2

Metric	Value
Outliers After	2

Screenshot



Automated EDA Report

```
ProfileReport()
```

Generated:

```
EDA_Report.html
```

Screenshot



Export Final Dataset

```
final_df.to_csv(
    "Final_Cleaned_Dataset.csv",
    index=False
)
```

Output

```
Dataset Saved Successfully!
```



Project Structure

```
Customer_Data_Project/
|
```

```
|— users.csv
|— users.db
|— user_dataset.json
|— Final_Cleaned_Dataset.csv
|— EDA_Report.html
|
|— notebooks/
|   └─ Data_Transformation.ipynb
|
|— screenshots/
|   └─ load_dataset.png
|   └─ dataset_info.png
|   └─ missing_values.png
|   └─ encoding.png
|   └─ scaling.png
|   └─ final_report.png
|
└─ README.md
```

Project Outcome

- ✓ Data Cleaning Completed
 - ✓ Missing Value Handling Completed
 - ✓ Outlier Detection Completed
 - ✓ Data Transformation Completed
 - ✓ Feature Engineering Completed
 - ✓ Feature Scaling Completed
 - ✓ EDA Report Generated
 - ✓ Final Dataset Exported
 - ✓ Machine Learning Ready Dataset Prepared
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Author

Ajay Zala

Data Science | Machine Learning | Python Developer

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